Creating and benchmarking Learned DB Components using LLM

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Agenda

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5. Active Learning
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Why Learned Cardinality Estimation?

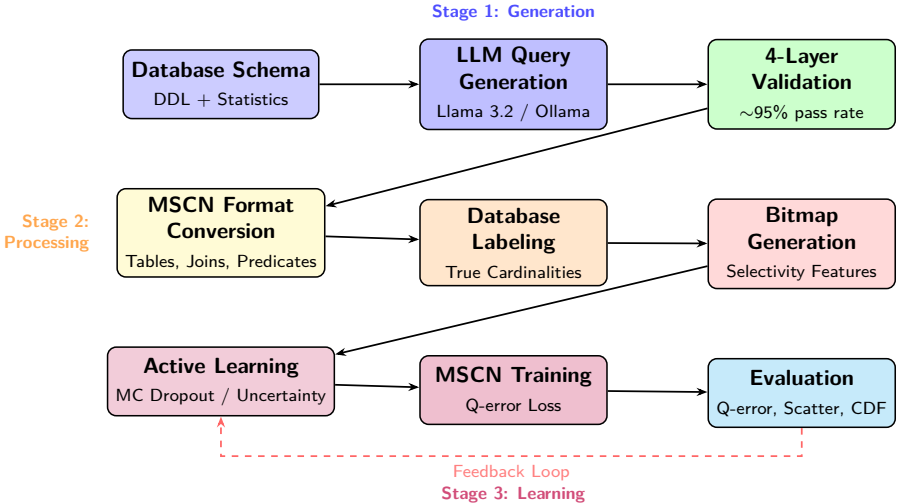
The Problem

- Bad cardinality estimates lead to catastrophic join orders
- Histograms and independence assumptions fail on correlated data
- Sampling is expensive at runtime and misses rare groups

The Learned Approach

- Deep learning models (MSCN) learn correlations directly from data
- Outperform traditional mathematical estimators
- **But:** Require massive labeled workloads (queries + true cardinalities)
- Obtaining labels via query execution is expensive and hard to scale

Pipeline: From Schema to Trained Estimator



LLM-based Query Generation

Schema-aware prompting with diversity control

Prompt Strategy:

- **Input:** Schema DDL + column statistics
- **Output:** JSON array of SQL queries
- **Diversity hints:** Rotate across 5 structural profiles
- **Batched generation:** Manages token limits, retries

4-Layer Validation:

- Keyword → Parse → Schema → Format
- ∼95% pass rate

Strict Constraints:

- `SELECT COUNT(*)` only
- 1–3 tables with proper aliases
- Joins on defined foreign keys
- Simple comparisons (=, <, >) with `AND`
- Prohibited: `OR`, `IN`, `LIKE`, subqueries

MSCN

Set-based encoding with masked pooling

Query Encoding:

- **Tables:** One-hot + bitmap sample
- **Predicates:** Column + operator + normalized value
- **Joins:** One-hot of join condition

Architecture:

- Parallel 2-layer MLPs per set
- Masked mean pooling (ignores padding)
- Concatenate → output MLP → sigmoid
- Loss: Mean Q-error (scale-invariant)

Active Learning for Cardinality Estimation

The Labeling Bottleneck: Every label requires executing a query on a live database. Complex joins can take seconds each. Labeling thousands of queries takes hours.

Standard Supervised

1. Generate massive dataset
2. **Label everything**
3. Train model once

Wasteful: labels redundant, easy queries

Active Learning

1. Label small seed (20%)
2. Train initial model
3. Score unlabeled queries by uncertainty
4. **Label only the hardest**
5. Retrain and repeat

Efficient: focuses budget where it matters

Acquisition Strategies

How to decide which queries to label next

1. **Random Sampling** — Baseline. No scoring, just diversity from the pool.
2. **Uncertainty Sampling** — Select queries where sigmoid output is closest to 0.5:

$$\alpha_{\text{uncertainty}}(q) = -\left|f_\theta(q) - 0.5\right|$$

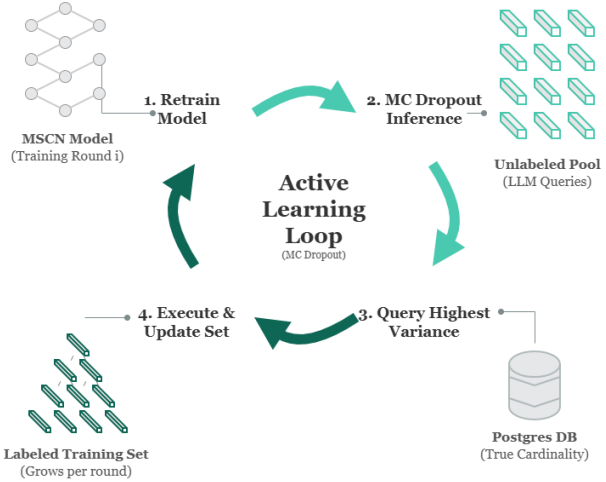
Cheap (single forward pass), but ignores diversity among selected queries.

3. **Monte Carlo Dropout** — Keep dropout active, run $T = 25$ forward passes:

$$\alpha_{\text{MC}}(q) = \text{Var}\left(\{\hat{y}^{(t)}\}_{t=1}^{T}\right)$$

High variance = fragile representation = most informative to label. Richer signal than plain uncertainty; modest overhead vs. database execution cost.

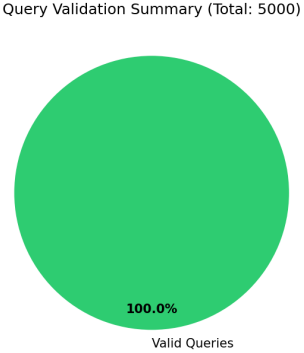
The Active Learning Loop



Experimental Setup & Data Quality

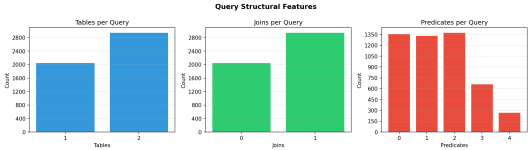
Configuration:

- **Database:** IMDB benchmark
- **LLM:** Llama 3.2 via Ollama
- **Model:** MSCN (hidden=256, dropout=0.1)
- **AL:** MC Dropout, $T = 25$, 5 rounds
- **Metric:** Q-error

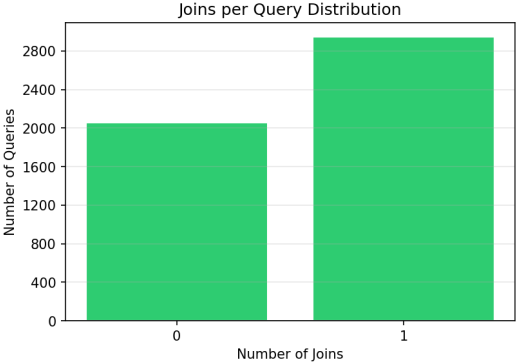


Query validation success rate across generation batches

Workload Diversity



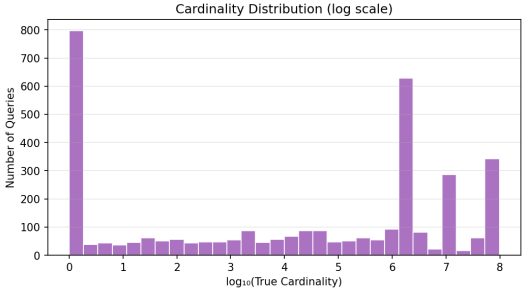
Structural feature distributions



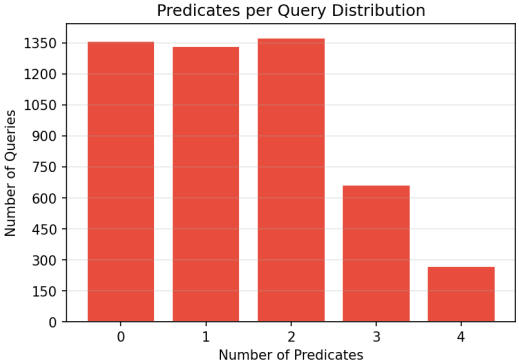
Join count distribution

Diversity hints produce a balanced mix of single-table, two-table, and three-table queries with varied predicate counts.

Cardinality & Predicate Coverage



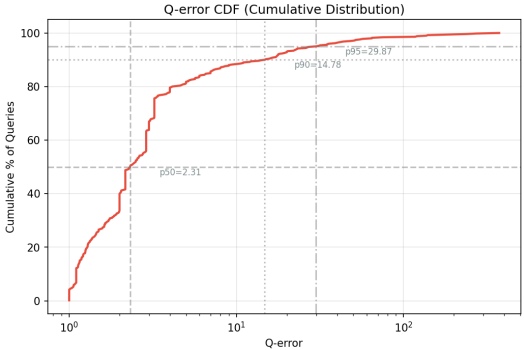
True cardinality distribution (log scale)



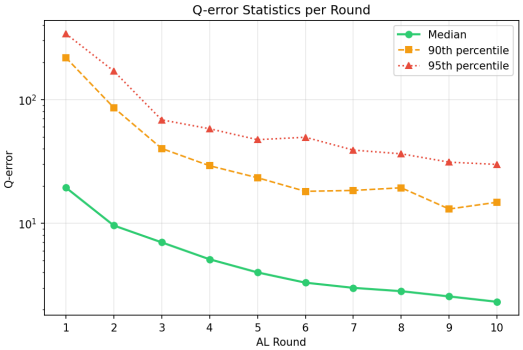
Predicate count distribution

Generated workload spans cardinalities from single-digit to millions, covering the full range needed for robust estimation.

Estimation Accuracy: Q-error Analysis



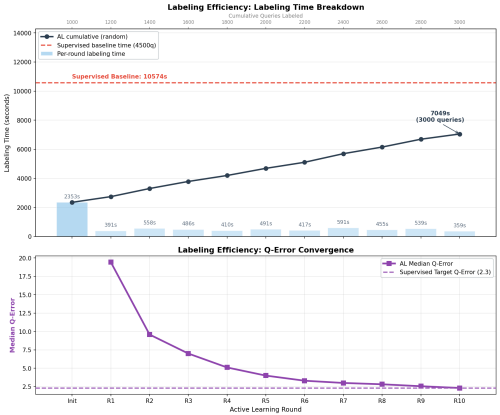
Q-error CDF: majority within factor of 2–5



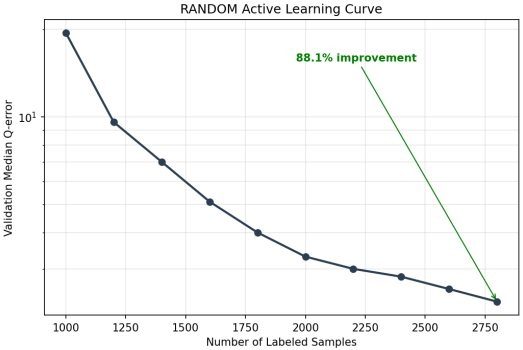
Q-error statistics per AL round

Median, mean, and tail Q-errors all decrease with each active learning round.

Active Learning: Labeling Efficiency



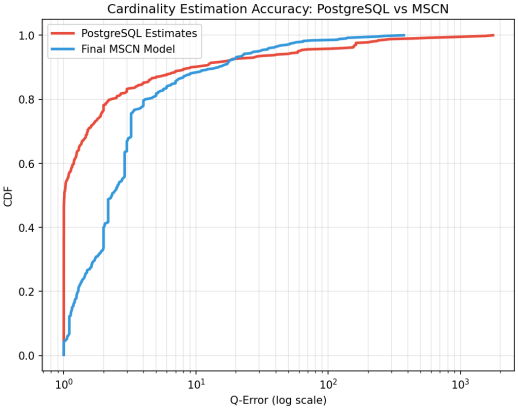
AL achieves competitive accuracy with fewer labels



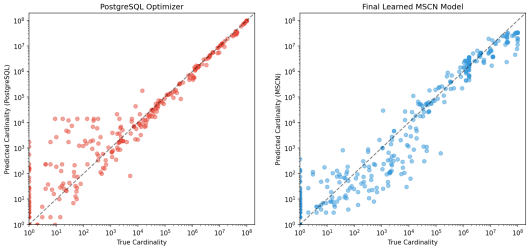
Learning curve across AL rounds

Intelligent query selection reduces labeling cost by ∼40% while maintaining estimation quality.

MSCN vs. PostgreSQL Optimizer



Q-error CDF: MSCN vs. PostgreSQL



Predicted vs. actual: closer to diagonal = better

The MSCN model trained with active learning matches or outperforms PostgreSQL's built-in estimator.

Conclusion

What we showed:

- LLMs generate diverse, realistic SQL workloads from schema alone ($\sim$95% valid)
- Active learning with MC Dropout reduces labeling cost by $\sim$40%
- The trained MSCN matches PostgreSQL's built-in estimator
- The pipeline is schema-agnostic and fully automated

Future Directions:

- Extend to richer SQL fragments (OR, LIKE, GROUP BY)
- Cross-schema transfer learning
- Integration with live query optimizers
- Hybrid acquisition: uncertainty + diversity

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Thank you

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